

5G/6G Communication Systems- ECA5604 (Design Task)

Student Name: U.Guru Prasad Reddy

Reg no: 192312683L

Experiment : Design a remote patient monitoring communication architecture.

Aim: To design a remote patient monitoring communication architecture.

Objective :

```
clc;
```

```
clear;
```

```
close all;
```

```
% Remote Patient Monitoring
```

```
t = 0:1:100;
```

```
% Patient sensor data
```

```
HR = 70 + 5*randn(size(t));    % Heart rate
```

```
SpO2 = 98 + randn(size(t));    % Oxygen level
```

```
Temp = 36.8 + 0.2*randn(size(t)); % Temperature
```

```
% Wireless communication
```

```
packetLoss = rand(size(t)) < 0.05;
```

```
HR(packetLoss) = NaN;
```

```
SpO2(packetLoss) = NaN;
```

```
Temp(packetLoss) = NaN;
```

```
% Alert detection
```

```
alert = HR > 100 | SpO2 < 92 | Temp > 38;
```

```
% Display
```

```
figure;
```

```
subplot(3,1,1)
plot(t,HR,'b'); yline(100,'r--');
ylabel('Heart Rate'); grid on;
```

```
subplot(3,1,2)
plot(t,SpO2,'g'); yline(92,'r--');
ylabel('SpO2 (%)'); grid on;
```

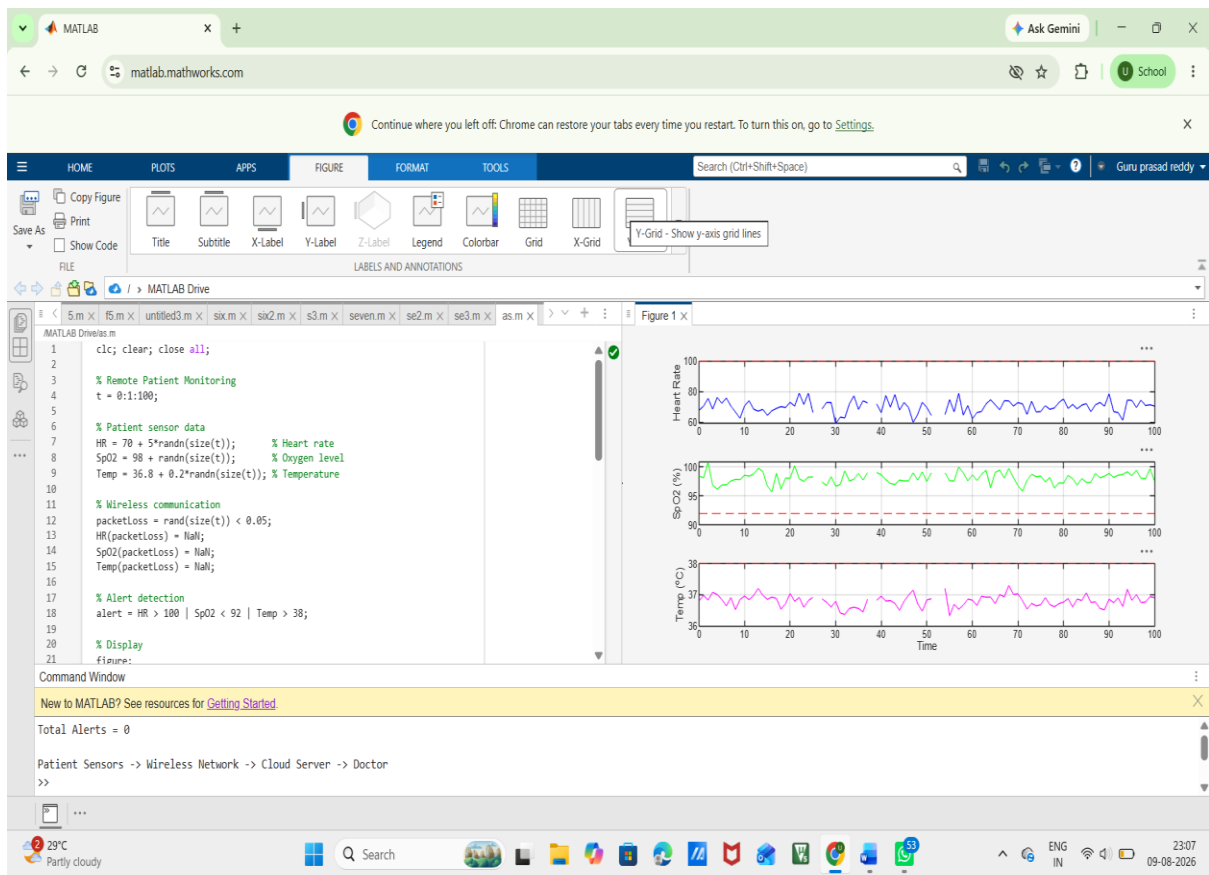
```
subplot(3,1,3)
plot(t,Temp,'m'); yline(38,'r--');
ylabel('Temp (^oC)'); xlabel('Time');
grid on;
```

```
fprintf('Total Alerts = %d\n',sum(alert));
```

```
% Architecture
```

```
fprintf('\nPatient Sensors -> Wireless Network -> Cloud Server -> Doctor\n');
```

Output:



Result: Thus, designed a remote patient monitoring communication architecture.

Experiment : Develop a wireless communication workflow for AI-assisted healthcare diagnosis.

Aim: To Develop a wireless communication workflow for AI-assisted healthcare diagnosis.

Objective :

clc;

clear;

close all;

% AI-Assisted Healthcare Wireless Communication

% Patient -> Sensors -> Wireless Network -> AI Server -> Doctor

t = 0:1:100;

% 1. Patient sensor data

HR = 75 + 5*randn(size(t)); % Heart rate

SpO2 = 97 + randn(size(t)); % Oxygen saturation

Temp = 36.8 + 0.2*randn(size(t)); % Temperature

% 2. Wireless transmission

packetLoss = rand(size(t)) < 0.05;

HR(packetLoss) = NaN;

SpO2(packetLoss) = NaN;

Temp(packetLoss) = NaN;

% 3. AI-assisted diagnosis

diagnosis = HR > 100 | SpO2 < 92 | Temp > 38;

% 4. Display results

figure;

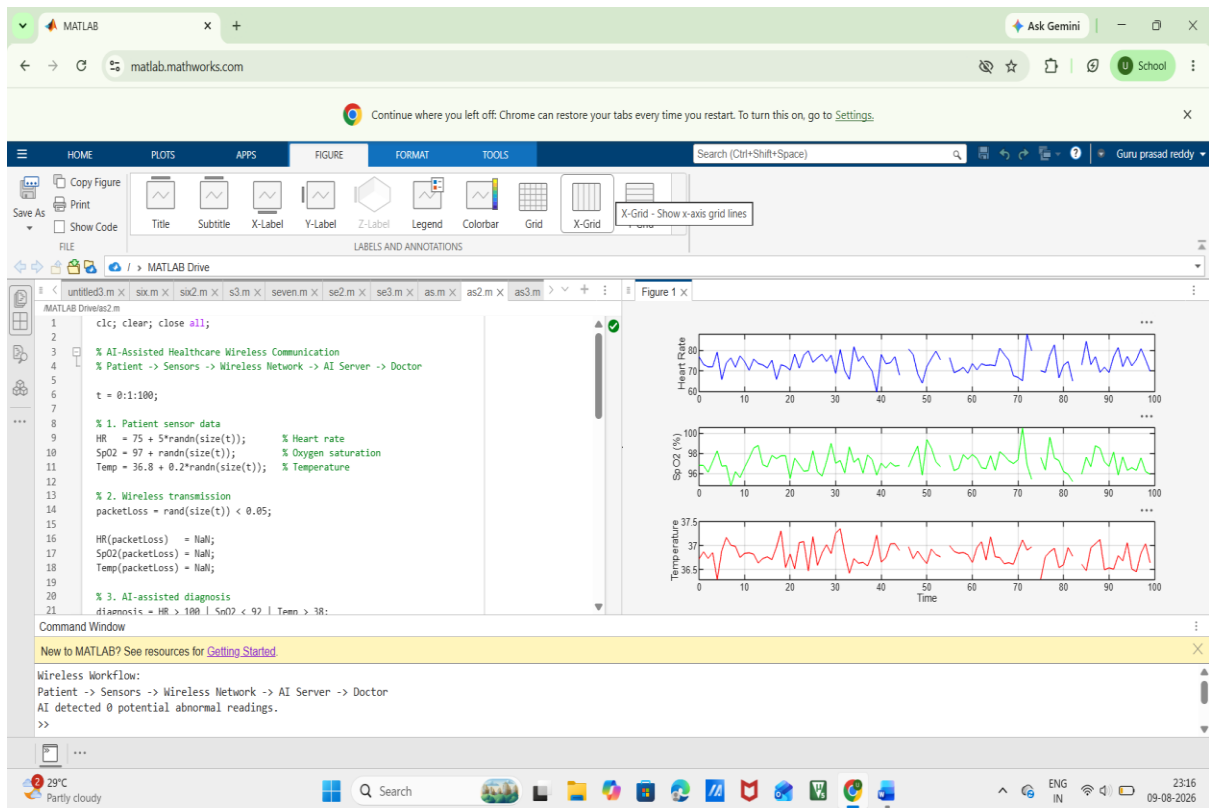
```
subplot(3,1,1);  
plot(t,HR,'b');  
ylabel('Heart Rate');  
grid on;
```

```
subplot(3,1,2);  
plot(t,SpO2,'g');  
ylabel('SpO2 (%)');  
grid on;
```

```
subplot(3,1,3);  
plot(t,Temp,'r');  
ylabel('Temperature');  
xlabel('Time');  
grid on;
```

```
fprintf('Wireless Workflow:\n');  
fprintf('Patient -> Sensors -> Wireless Network -> AI Server -> Doctor\n');  
fprintf('AI detected %d potential abnormal readings.\n',sum(diagnosis));
```

Output:



Result:

Thus, Developed a wireless communication workflow for AI-assisted healthcare diagnosis.